Note

- $A \times B$ is the set of all possible ordered pairs between the elements of A and B such that the first coordinate is an element of A and the second coordinate is an element of B.
- $B \times A$ is the set of all possible ordered pairs between the elements of A and B such that the first coordinate is an element of B and the second coordinate is an element of A.
- In general $(a, b) \neq (b, a)$, in particular, if $a = b$, then $(a, b) = (b, a)$.
- The "cartesian product" is also referred as "cross product".

Illustration 2

Let A = {1, 2, 3} and B = {a, b}. Write $A \times B$ and $B \times A$?

$A \times B = \{1,2,3\} \times \{a,b\} = \{(1,a),(1,b),(2,a),(2,b),(3,a),(3,b)\}$ (as shown in Fig.1.3)

$B \times A = \{a,b\} \times \{1,2,3\} = \{(a,1), (a,2), (a,3),(b,1), (b,2), (b,3)\}$ (as shown in Fig.1.3)

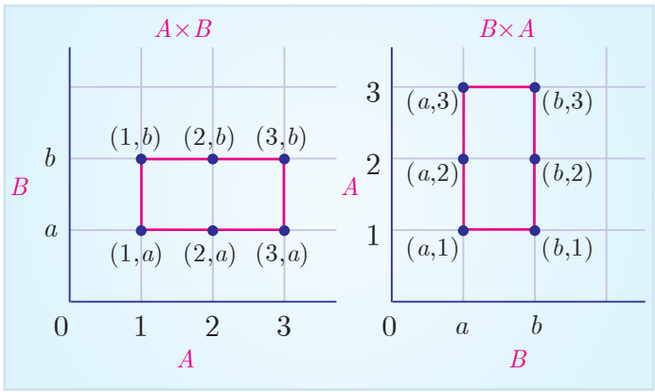


Fig. 1.3

Thinking Corner

When will $A \times B$ be equal to $B \times A$?

Note

- In general $A \times B \neq B \times A$, but $n(A \times B) = n(B \times A)$
- $A \times B = \phi$ if and only if $A = \phi$ or $B = \phi$
- If $n(A) = p$ and $n(B) = q$ then $n(A \times B) = pq$

Recall of standard infinite sets

Natural Numbers $\mathbb{N} = \{1,2,3,4\ldots\}$; Whole Numbers $\mathbb{W} = \{0,1,2,3, \ldots\}$;

Integers $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$; Rational Numbers $\mathbb{Q} = \left\{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\right\}$;

Real Numbers $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'$, where $\mathbb{Q}'$ is the set of all irrational numbers.

Illustration 3

For example, let A be the set of numbers in the interval [3, 5] and B be the set of numbers in the interval [2,3]. Then the Cartesian product $A \times B$ corresponds to the rectangular region shown in the Fig. 1.4. It consists of all points (x, y) within the region.

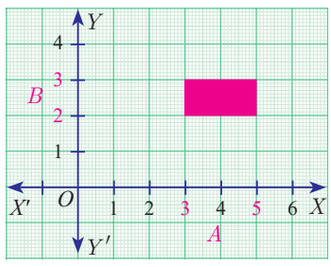


Fig. 1.4

Progress Check

1. For any two non-empty sets A and B, $A \times B$ is called as _____.
2. If $n(A \times B) = 20$ and $n(A) = 5$ then $n(B)$ is _____.
3. If $A = \{-1,1\}$ and $B = \{-1,1\}$ then geometrically describe the set of points of $A \times B$.
4. If A, B are the line segments given by the intervals (–4, 3) and (–2, 3) respectively, represent the cartesian product of A and B.

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